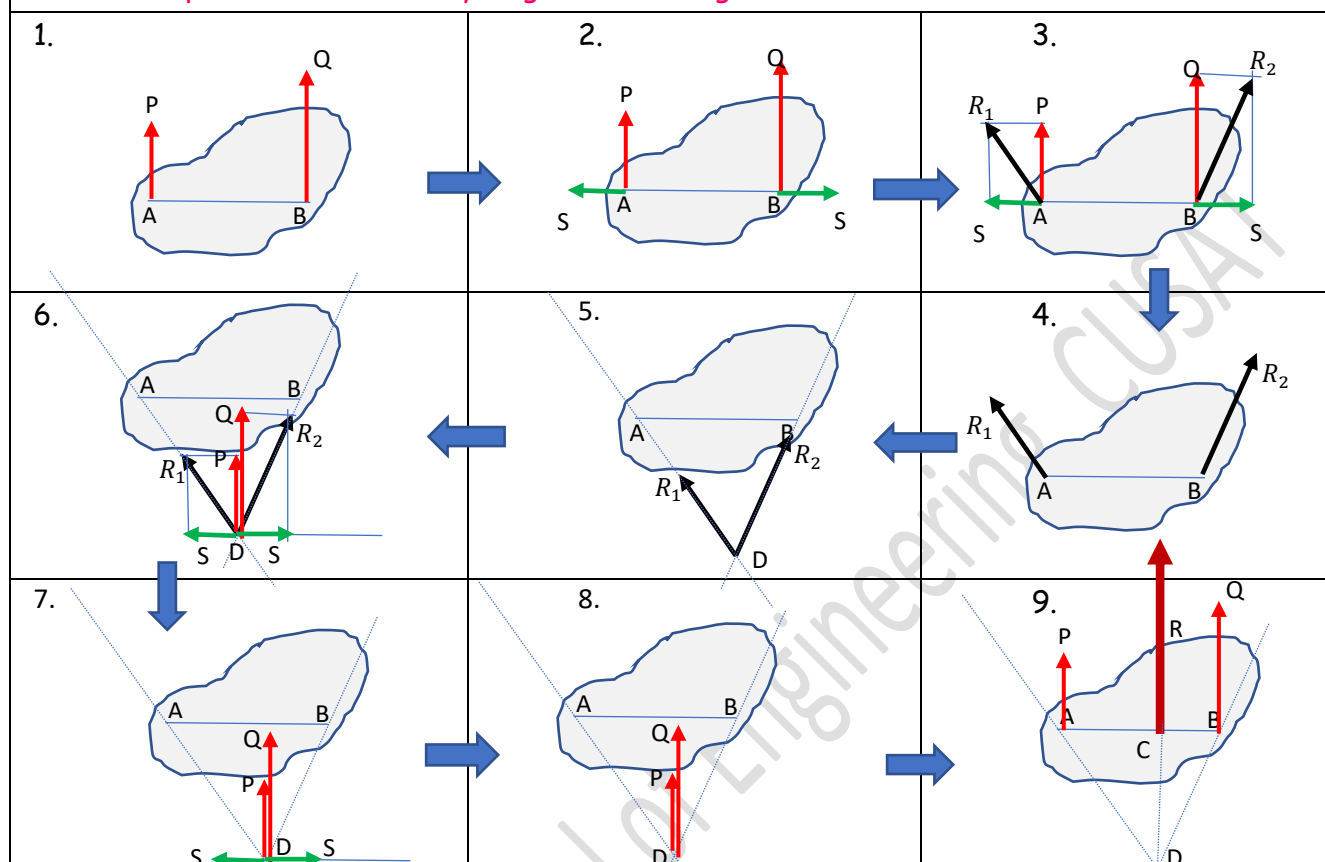


Parallel forces in a plane

After considering systems of concurrent forces, let us now consider systems of parallel forces. We start with two parallel forces and see how to find their resultant.

1. Two parallel forces of any magnitudes acting in the same direction



1. Consider two parallel forces of magnitude P and Q acting on a body. By applying the principle of transmissibility, these forces can be transmitted along their lines of action and brought to, say, points A and B .
2. To this, let us add two equal, opposite, and collinear forces of magnitude S (which are in equilibrium) at points A and B (Principle of superposition).
3. Let R_1 be the resultant of P and S at A and R_2 be the resultant of Q and S at B .
4. Now the given force system is replaced by two inclined forces R_1 and R_2 .
5. These two forces are transmitted along their lines to meet at point D .
6. Then, R_1 and R_2 are resolved back to P & S and Q & S respectively.
7. We have now four forces; P and Q acting in the same direction from D , S and S acting in the opposite directions from D .
8. The two equal, opposite, and collinear forces S and S which are in equilibrium can be removed from the system (Principle of superposition).
9. The resultant of the two forces P and Q acting from D in the same direction is given by the parallelogram law of forces; $R = P + Q$. The direction of R is in the direction of the forces P & Q . The line of action of R intersects the line AB at C .

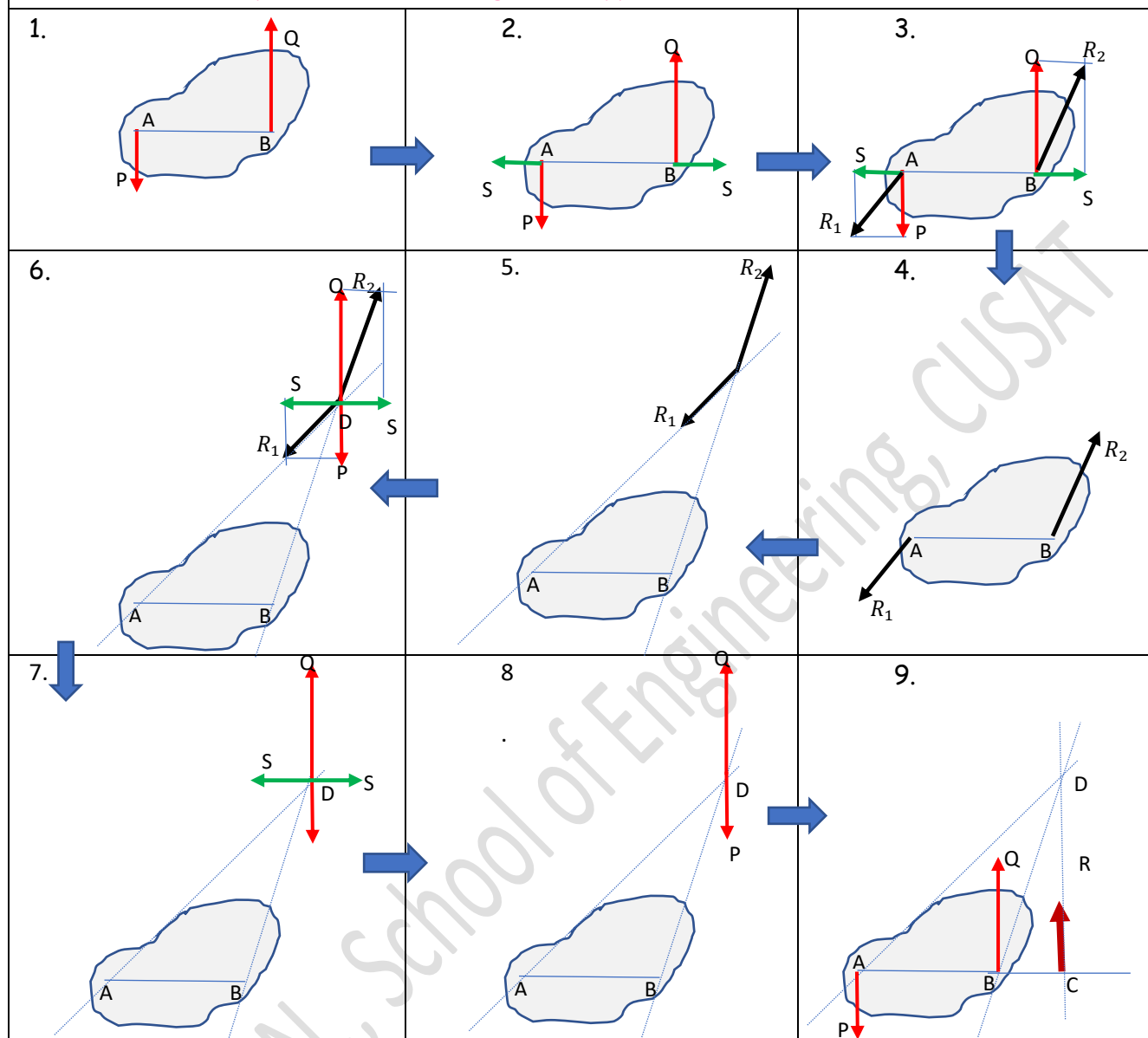
Applying Varignon's theorem;

moment of R about point A = algebraic sum of moments of P and Q about A .

$$+R \cdot \overline{AC} = 0 + Q \cdot \overline{AB} \gg (P + Q) \overline{AC} = Q \overline{AB} \gg \gg P \cdot \overline{AC} = Q \cdot (\overline{AB} - \overline{AC}) \gg \gg \frac{P}{Q} = \frac{\overline{BC}}{\overline{AC}}$$

The line of action of the resultant divides the line between P and Q internally in the inverse ratio of their magnitude. R is closer to the bigger force.

2. Two unequal parallel forces acting in the opposite direction



1. Consider two unequal parallel forces of magnitude P and Q ($Q > P$) acting on a body. By applying the principle of transmissibility, these forces can be transmitted along their lines of action and brought to, say, points A and B .
2. To this, let us add two equal, opposite, and collinear forces of magnitude S (which are in equilibrium) at points A and B (Principle of superposition).
3. Let R_1 be the resultant of P and S at A and R_2 be the resultant of Q and S at B .
4. Now the given force system is replaced by two inclined forces R_1 and R_2 .
5. These two forces are transmitted along their lines to meet at point D .
6. Then, R_1 and R_2 are resolved back to P & S and Q & S respectively.
7. We have now four forces; P and Q acting in the opposite direction from D , S and S acting in the opposite directions from D .
8. The two equal, opposite, and collinear forces S and S which are in equilibrium can be removed from the system (Principle of superposition).
9. The resultant of the two forces P and Q acting from D in the opposite direction is given by the parallelogram law of forces; $R = Q - P$. The direction of R is in the direction of the bigger force Q . The line of action of R intersects the line AB at C .

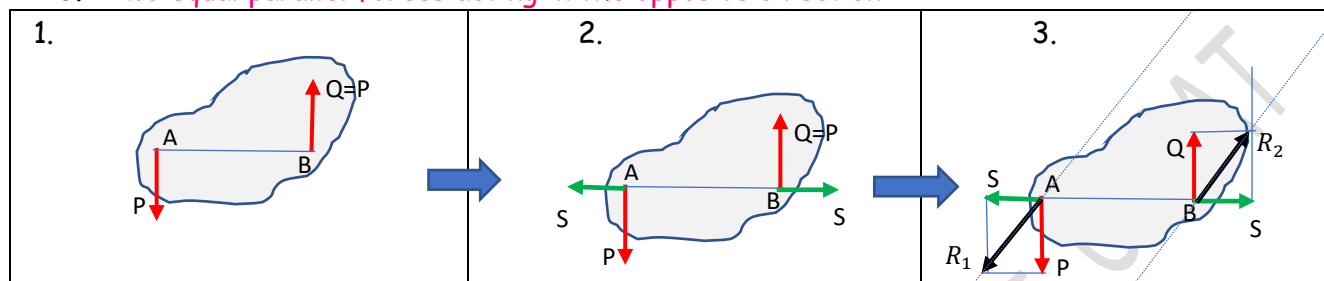
Applying Varignon's theorem;

moment of R about point A = algebraic sum of moments of P and Q about A.

$$+R \cdot \overline{AC} = 0 + Q \cdot \overline{AB} \gg (Q - P)\overline{AC} = Q \overline{AB} \gg \gg P \cdot \overline{AC} = Q \cdot (\overline{AC} - \overline{AB}) \gg \gg \frac{P}{Q} = \frac{\overline{BC}}{\overline{AC}}$$

The line of action of the resultant divides the line between P and Q externally in the inverse ratio of their magnitude. R is passing through a point outside the space between P and Q and on the side of the bigger force.

3. Two equal parallel forces acting in the opposite direction

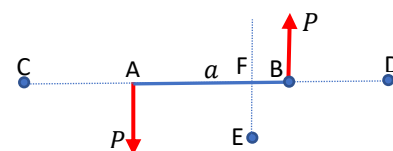


1. Consider two equal parallel forces of magnitude P and Q ($Q = P$) acting on a body. By applying the principle of transmissibility, these forces can be transmitted along their lines of action and brought to, say, points A and B .
2. To this, let us add two equal, opposite, and collinear forces of magnitude S (which are in equilibrium) at points A and B (Principle of superposition).
3. Let R_1 be the resultant of P and S at A and R_2 be the resultant of Q and S at B .
4. Now the given force system is replaced by two inclined forces R_1 and R_2 .
5. These two forces are transmitted along their lines. They will never intersect as their lines of action are parallel.
6. The two equal, opposite, and parallel forces form a couple whose resultant force is equal to zero but will have a resultant moment.

Properties of couple

1. The moment of a couple is equal to the product of the magnitude of one of the forces (P) and arm of the couple (a). The moment of a couple is independent of the moment centre.

Moment of the couple, $M = +Pa$ (counter clockwise)



Moment of the forces about A, $M_A = +Pa$

Moment of the forces about B, $M_B = +Pa$

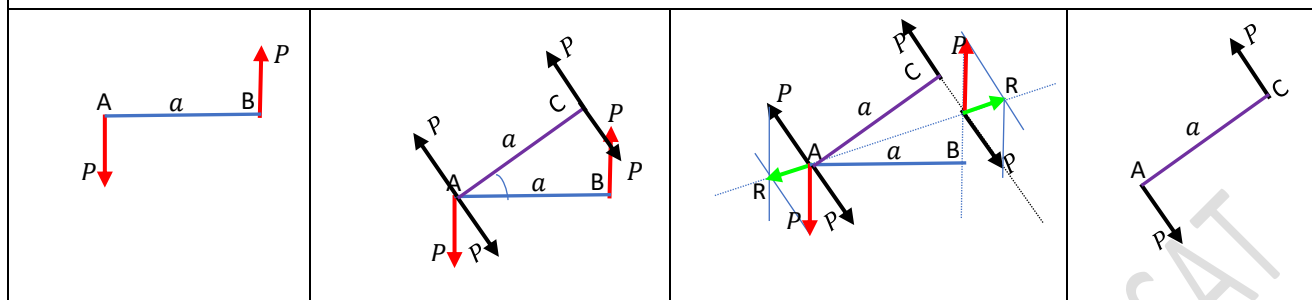
Moment of the forces about C, $M_C = +P \times BC - P \times AC = +P(BC - AC) = +Pa$

Moment of the forces about D, $M_D = +P \times AD - P \times BD = +P(AD - BD) = +Pa$

Moment of the forces about E, $M_E = +P \times AF + P \times BF = +P(AF + BF) = +Pa$

Hence, the moment of a couple is independent of the moment centre and is always equal to the product of one of the forces and the arm of the couple.

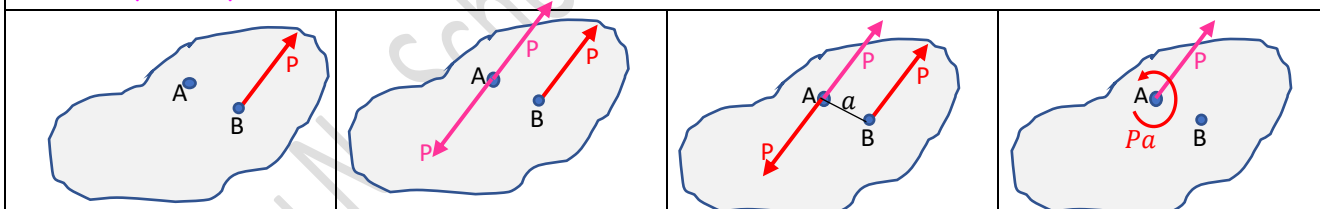
2. A couple can be moved anywhere in its plane without any change in the moment of the couple



- Consider a couple formed by two equal, opposite, and parallel forces P and P acting at A and B . The moment of the couple is equal to $+Pa$ (counter clockwise).
- Consider another line AC at some angle whose length is equal to AB . Add two equal, opposite, and collinear forces P and P at A and B perpendicular to AC (represented by black vectors). This addition will not produce any change as per the principle of superposition.
- The force at B and one force at C are transmitted along their lines of action to a common point and their resultant R is found out using parallelogram law. The resultant R' of two forces (one red and one black) at A is also found out. These resultant R and R' are equal, opposite, and collinear. They are in equilibrium and can be removed from the system.
- Now, we are left with two equal, opposite, and parallel force P and P at A and C . They form a couple of magnitude $+Pa$.

Thus, we have turned the original couple about one of its ends. By rotating the couple about its ends successively, we can move the couple anywhere in its plane.

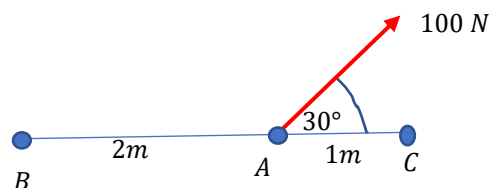
3. When a force is transferred to any point not in its line of action, it will be accompanied by a couple.



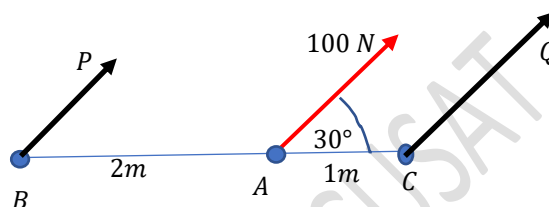
Suppose, we would like to transfer a force P acting at point B in a body to point A which is not in its line of action. Add two equal, opposite, and collinear forces P and P at A whose line of action is parallel to that of the original force. By the principle of superposition, this addition will not create any change in the action of the original force. Now, the original force P at B and the parallel opposite force P at A form a couple whose moment is equal to Pa counter clockwise. Thus when a force is transferred to another point not in its line of action, it will be accompanied by a couple.

Example 1

Resolve the force 100 N in to two forces at B and C



Since the force is to be resolved in to two forces on either side of it, the components should be parallel to the given force and should be in the same direction as that of the given force. The resultant should be closer to the bigger force



$$100\text{ N} = P + Q$$

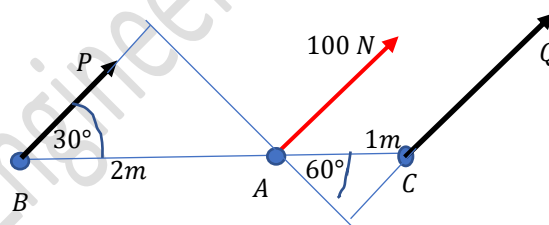
$$\frac{P}{Q} = \frac{AC}{AB} = \frac{1}{2} \gggg 2P = Q$$

$$100\text{ N} = P + 2P \gggg P = 33.3\text{ N} \gggg Q = 66.7\text{ N}$$

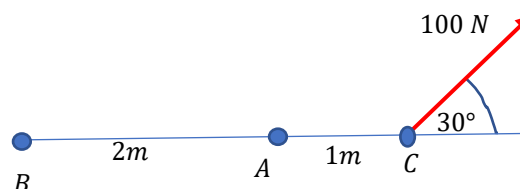
We can also use Varignon's theorem. Moment of resultant about A is equal to algebraic sum of moments of components about A.

$$0 = Q \times 1 \cos 60 - P \times 2 \sin 30$$

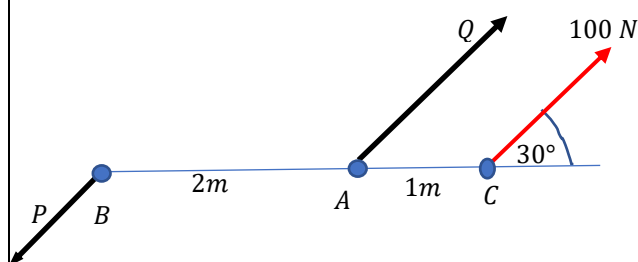
$$Q \times \frac{1}{2} = P \times 2 \times \frac{1}{2} \gggg Q = 2P$$

**Example 2**

Resolve the force 100 N in to two forces at A and B



Since the force is to be resolved in to two forces on the same side of it, the components should be parallel to the given force and should be in the opposite directions. The bigger force should be closer to the resultant and should be in the same direction of the resultant.



$$100\text{ N} = Q - P$$

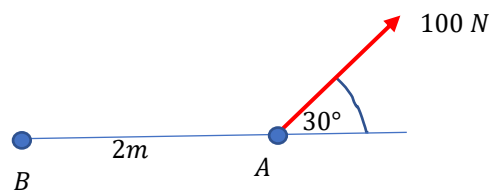
$$\frac{P}{Q} = \frac{AC}{BC} = \frac{1}{3} \gggg 3P = Q$$

$$100\text{ N} = 3P - P \gggg P = 50\text{ N}$$

$$\gggg Q = 150\text{ N}$$

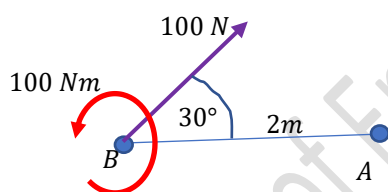
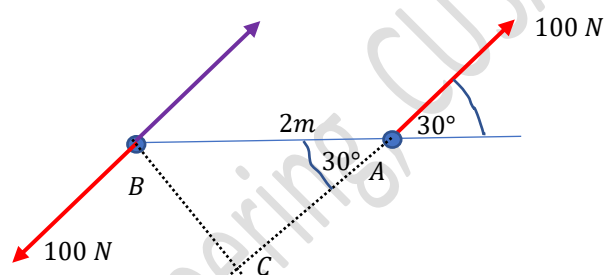
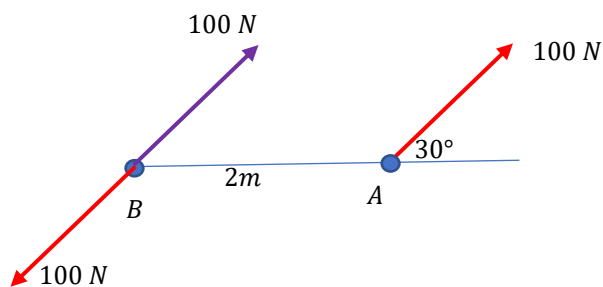
Example 3

Transfer the force 100 N to point B.

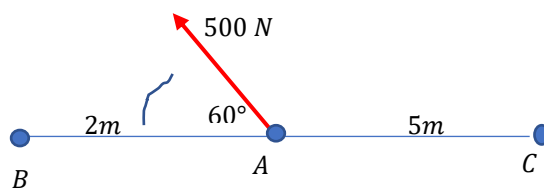


Add two equal, opposite, and collinear forces 100 N at B parallel to the given force. The equal, opposite, and parallel forces 100 N (marked by red) form a couple whose magnitude is given by

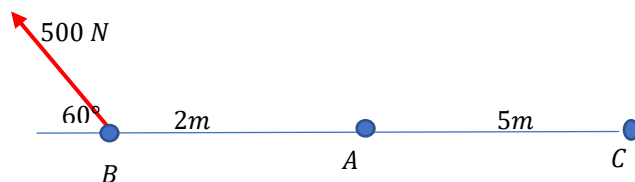
$$M = +100 \times 2 \sin 30 = +100 \text{ Nm}$$

**Exercise 8****8.1**

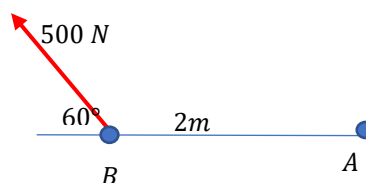
Resolve the force 500 N in to two forces at B and C

**8.2**

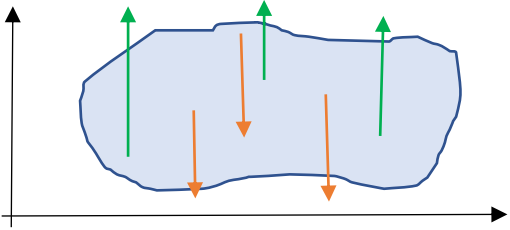
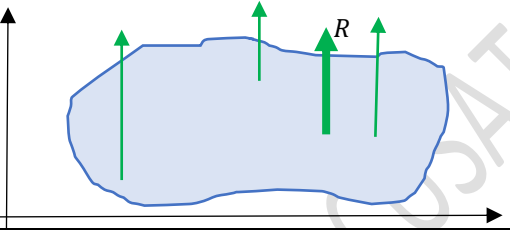
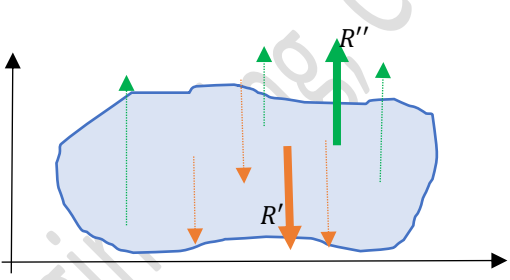
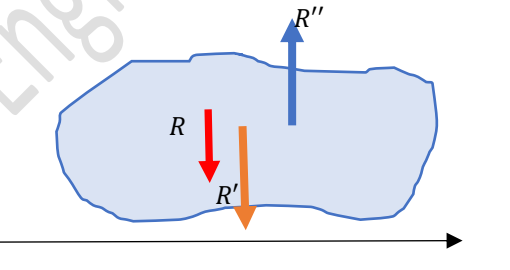
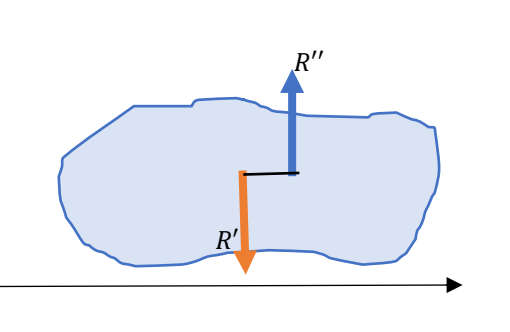
Resolve the force 500 N in to two forces at A and C

**8.3**

Transfer the force 500 N to point A.

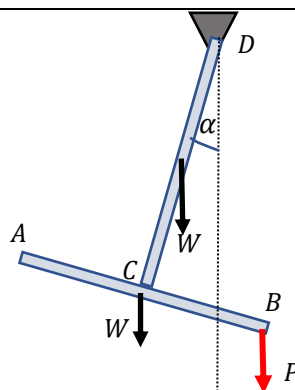


Equilibrium of a system of parallel forces

When there are a number of forces which are parallel to each other, we have a system of parallel forces. We can consider two forces at a time and reduce the forces in to only two parallel forces finally.	 A diagram showing a light blue irregularly shaped body on a horizontal axis. Several green arrows point upwards and several orange arrows point downwards, representing a system of parallel forces.
i) If all the forces are in the same direction, there will be a resultant force;	 A diagram showing a light blue irregularly shaped body on a horizontal axis. All green arrows point upwards. A single green arrow labeled 'R' represents the resultant force.
ii) if we have forces in opposite directions, then we can reduce them in to two parallel forces R' and R'' acting in opposite directions as shown.	 A diagram showing a light blue irregularly shaped body on a horizontal axis. It shows two sets of parallel forces: one set of green arrows pointing up and one set of orange arrows pointing down. These are reduced to two single arrows: a green arrow labeled R'' pointing up and an orange arrow labeled R' pointing down.
a) If R' and R'' are unequal and parallel, we can find out the resultant R of the two.	 A diagram showing a light blue irregularly shaped body on a horizontal axis. It shows a blue arrow labeled R'' pointing up and an orange arrow labeled R' pointing down. A red arrow labeled R represents the resultant of these two forces.
b) If R' and R'' are equal the resultant force is zero irrespective of their line of action. But, if the lines of action are different, then they form a couple with moment M Hence, a system of parallel forces will be in equilibrium only if $\Sigma F = 0$ and $\Sigma M = 0$	 A diagram showing a light blue irregularly shaped body on a horizontal axis. It shows a blue arrow labeled R'' pointing up and an orange arrow labeled R' pointing down. The arrows are of equal length and are separated by a distance, representing a couple.

Example 1

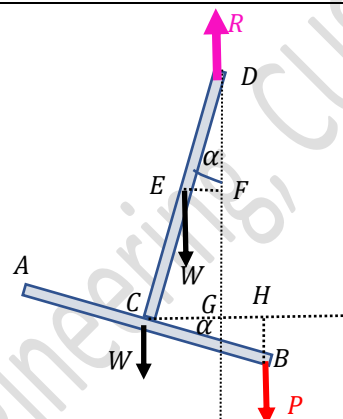
Two identical prismatic bars AB and CD of length $l = 1\text{m}$ and weight $W = 25\text{N}$ are joined to form a T-bar and suspended in a vertical plane as shown. Calculate the angle α that the bar CD will make with the vertical for equilibrium when a load of $P = 50\text{N}$ is suspended from the end B .



Consider the free body diagram of the T-bar. As there is no external horizontal forces acting on the system, the reaction at D is having only vertical component R (Even if we take a horizontal component, it will be equal to zero when we use $\Sigma F_x = 0$). The forces now form a system of parallel forces. For equilibrium,

$$\Sigma F_y = 0 \gg R - 2W - P = 0 \gg R = 100\text{ N}$$

$$\Sigma M_D = 0 \gg W \times EF + W \times CG - P \times GH = 0$$



$$EF = ED \sin \alpha = \frac{l}{2} \sin \alpha \ll \gg CG = CD \sin \alpha = l \sin \alpha \ll \gg GH = CH - CG = CB \cos \alpha - CD \sin \alpha$$

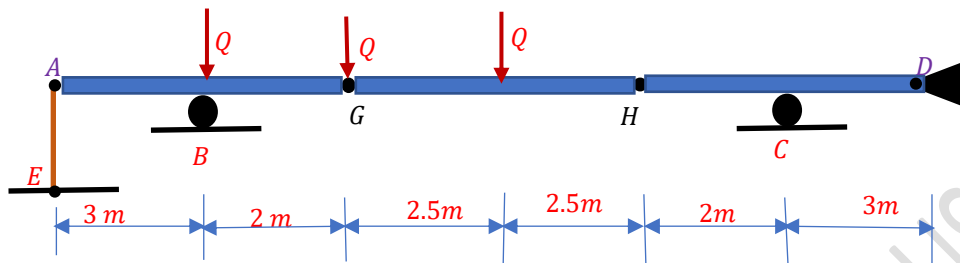
$$\therefore 25 \times \frac{l}{2} \sin \alpha + 25 \times l \sin \alpha = 50 \times \left(\frac{l}{2} \cos \alpha - l \sin \alpha \right)$$

$$\sin \alpha \left(\frac{1}{2} + 1 + 2 \right) = \cos \alpha$$

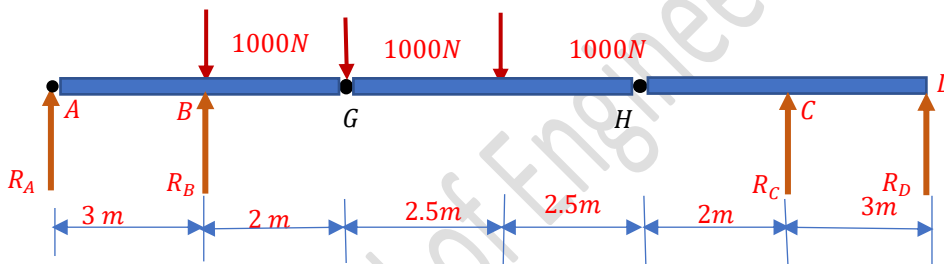
$$\tan \alpha = \frac{2}{7} \gg \gg \gg \alpha = 16^\circ$$

Example 2

Three beams are hinged together at their ends and loaded as shown in the figure given below. The end A is supported on a rod AE, B and C on rollers, and end D is hinged. If the force $Q = 1000\text{N}$, determine the support reactions at A, B, C, and D.



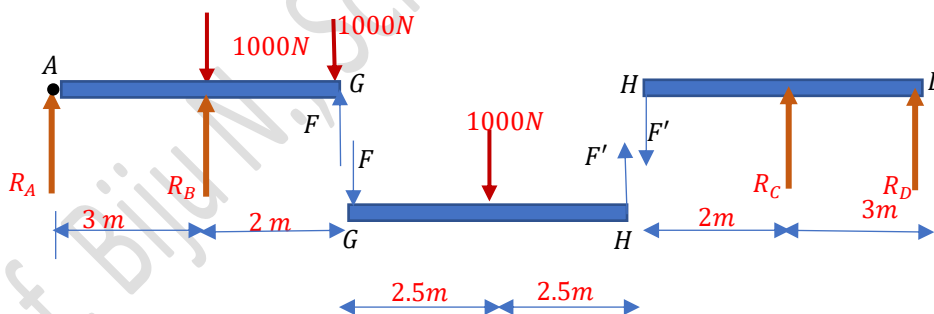
The end A is supported on a rod, which is a two-force member. The reaction at A is along the axis of the rod. The reactions at B and C are also vertical, as they are roller supports. The reaction at the hinge D will also be vertical, as there are no horizontal forces in the system. Thus, all the forces are in the vertical direction and they form a system of parallel forces. The free body diagram of the complete system is shown below.



$$\Sigma F_y = 0 \gg R_A + R_B + R_C + R_D - 1000 - 1000 - 1000 = 0$$

$$\Sigma M_A = 0 \gg R_B \times 3 + R_C \times 12 + R_D \times 15 - 1000 \times 3 - 1000 \times 5 - 1000 \times 7.5 = 0$$

$$R_B + 4R_C + 5R_D = 5166.7$$



Equilibrium of GH

$$\Sigma M_G = 0$$

$$F' \times 5 - 1000 \times 2.5 = 0$$

$$F' = 500\text{N}$$

$$\Sigma F_y = 0$$

$$F' - F - 1000 = 0$$

$$F = -500\text{N}$$

Equilibrium of AG

$$\Sigma M_A = 0$$

$$F \times 5 + R_B \times 3 - 1000 \times 3 - 1000 \times 5 = 0$$

$$R_B = 3500\text{N}$$

$$\Sigma F_y = 0 \gg R_A + R_B + F - 1000 - 1000 = 0$$

$$R_A = 1000\text{N}$$

Equilibrium of HD

$$\Sigma M_D = 0$$

$$F' \times 5 - R_C \times 3 = 0$$

$$R_C = 833.3\text{N}$$

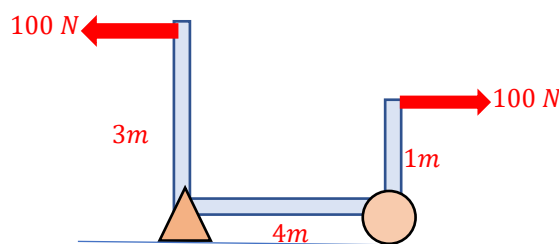
$$\Sigma F_y = 0$$

$$R_C + R_D - F' = 0$$

$$R_D = -333.3\text{N}$$

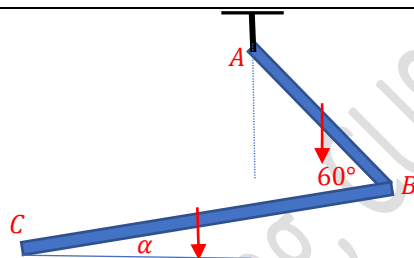
Exercise 9

9.1 Determine the support reactions for the structure loaded and supported as shown.



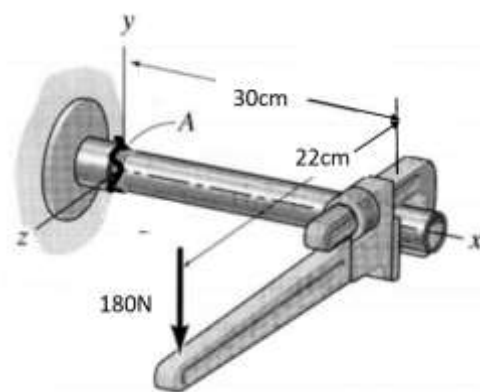
9.2

Two prismatic bars AB and BC of lengths 1m and 2m and weights 100 N and 200 N respectively are rigidly joined at B and suspended by a string as shown. Determine the position of equilibrium as defined by the angle α that the bars will assume.



9.3

A vertical force $P = 180\text{ N}$ is applied to the handle of a pipe wrench, whose handle is parallel to the z -axis. What is its effect on the base A of the pipe?



Reference

7. Engineering Mechanics: Timoshenko and Young, McGraw Hill
8. Mechanics for Engineers: Beer & Johnston, McGraw Hill
9. Engineering Mechanics: Merriam & Kraig, Wiley
10. Engineering Mechanics: Hibbeler, Pearson

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